Quantum Data Encoding and Feature Maps

EVA: Quantum Machine Learning – Lecture 5

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Roadmap

1. Why encoding is the bottleneck in QML
2. Basis, angle, and amplitude encoding
3. Feature maps as unitaries and kernels as overlaps
4. Entangling feature maps, kernel estimation, and re-uploading

Bridge from Lecture 4

In Lecture 4, kernels changed geometry without explicitly listing all coordinates.

Today the same idea appears in Hilbert space:

$$x \longmapsto |\phi(x)\rangle = U_\phi(x)\,|0\rangle^{\otimes n}, \qquad K_Q(x,y) = |\langle\phi(x)|\phi(y)\rangle|^2.$$

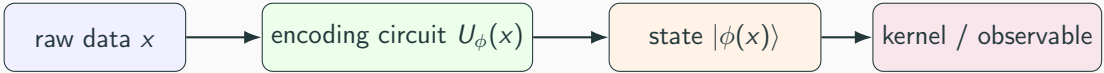
The encoding circuit decides what information is exposed to the model.

Why data encoding is a bottleneck

Classical data arrives as a vector

$$x = (x_1, \ldots, x_d) \in \mathbb{R}^d.$$

A quantum model does not act on that vector directly. It acts on a state prepared from the vector.



A poor encoding can destroy useful structure before learning even begins.

Recorded pause checkpoints

In the recorded version of this lecture, some slides are pause points for a short card set.

The five checkpoints are:

- Pause A: encoding basics
- Pause B: amplitude and angle calculations
- Pause C: overlap and fidelity
- Pause D: entangling feature maps
- Pause E: kernel estimation workflow

Data encoding strategies

Basis encoding

A binary string is mapped directly to a computational-basis state:

$$00 \mapsto |00\rangle\,, \qquad 01 \mapsto |01\rangle\,, \qquad 10 \mapsto |10\rangle\,.$$

Strengths:

- conceptually simple
- natural for discrete or already-binarized data

Limitations:

- real-valued features need preprocessing or discretization
- similarity structure is usually very rigid

Angle encoding

Each qubit starts in $|0\rangle$ by default, then features enter through rotation angles, for example

$$R_y(x_i) = \begin{bmatrix} \cos(x_i/2) & -\sin(x_i/2) \\ \sin(x_i/2) & \cos(x_i/2) \end{bmatrix}, \qquad R_z(x_i) = \begin{bmatrix} e^{-ix_i/2} & 0 \\ 0 & e^{ix_i/2} \end{bmatrix}.$$

Common pattern:

$$|\psi(x)\rangle = \bigotimes_{i=1}^{n} R_y(x_i)\,|0\rangle\,.$$

Important implication:

- no unit-norm constraint on raw features
- periodic gates imply aliasing: x_i and $x_i + 4\pi$ give the same one-qubit state under R_y
- scaling still matters, because it controls how fast phases/amplitudes change with feature values

Amplitude encoding

A normalized vector becomes amplitudes of a quantum state:

$$x = (x_0, \ldots, x_{2^n-1}) \quad \Longrightarrow \quad |\psi(x)\rangle = \sum_{j=0}^{2^n-1} x_j \, |j\rangle \,, \qquad \sum_j |x_j|^2 = 1.$$

Strengths:

- 2^n amplitudes fit into n qubits
- compact representation in principle

Limitations:

- normalization is mandatory
- generic state preparation can be deep

Compact in representation does not automatically mean cheap in circuit depth.

`lecture_05_demo.ipynb`, Demo 1: basis, amplitude, and angle encoding states.

Encoding strategies at a glance

Encoding	Main idea	Strength	Main caveat
Basis	bitstring $\to \lvert z \rangle$	simple and discrete	poor for continuous data
Amplitude	values $\to$ amplitudes	compact storage	state preparation cost
Angle	values $\to$ rotations	easy circuit construction	geometry depends on gate choice

How many real features can two qubits carry?

A pure n-qubit state has

$$2^{n+1} - 2$$

real degrees of freedom after normalization and global-phase removal.

For $n = 2$, this is 6.

So in principle two qubits can represent more than three real parameters. The practical limit is imposed by the chosen circuit ansatz, not by Hilbert space alone.

Example: a product map

$$R_y(x_1) \otimes R_y(x_2) \, |00\rangle$$

directly injects only two scalars, even though the full two-qubit state space is six-dimensional.

Pause A: solve Card Set A

Worked amplitude-encoding example

Suppose

$$x = (0.3, 0.7, 0.5, 0.1).$$

Its norm is

$$\|x\| = \sqrt{0.3^2 + 0.7^2 + 0.5^2 + 0.1^2} = \sqrt{0.84}.$$

Therefore the normalized state is

$$|\psi(x)\rangle = \frac{1}{\sqrt{0.84}} \left(0.3\,|00\rangle + 0.7\,|01\rangle + 0.5\,|10\rangle + 0.1\,|11\rangle\right).$$

`lecture_05_demo.ipynb`, Demo 1: explicit amplitude-encoding example.

How do we synthesize an amplitude-encoding circuit?

Goal:

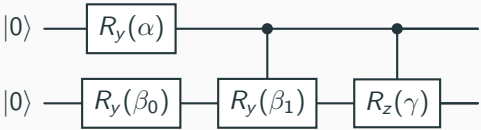
$$|\psi\rangle = \sum_{j=0}^{2^n-1} a_j \, |j\rangle \, , \qquad \sum_j |a_j|^2 = 1.$$

For two qubits, a constructive scheme is:

- choose one rotation on qubit 0 to split total probability between upper and lower branches
- use conditional rotations on qubit 1 to set branch-wise amplitude ratios
- if amplitudes are complex, add controlled phase gates

Two-qubit amplitude-synthesis template

Typical template:



where angles are solved from target amplitude ratios.

Message: amplitude encoding is constructive, but generic state preparation can still become deep as qubit count grows.

Worked angle-encoding example

Take two features and encode them by

$$|\psi(x)\rangle = (R_y(x_1)\,|0\rangle) \otimes (R_y(x_2)\,|0\rangle)\,.$$

Since

$$R_y(\theta)\,|0\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)|1\rangle\,,$$

the two-qubit state becomes

$$\begin{aligned}|\psi(x)\rangle = {} & \cos\left(\frac{x_1}{2}\right)\cos\left(\frac{x_2}{2}\right)|00\rangle + \cos\left(\frac{x_1}{2}\right)\sin\left(\frac{x_2}{2}\right)|01\rangle \\ & + \sin\left(\frac{x_1}{2}\right)\cos\left(\frac{x_2}{2}\right)|10\rangle + \sin\left(\frac{x_1}{2}\right)\sin\left(\frac{x_2}{2}\right)|11\rangle\,.\end{aligned}$$

`lecture_05_demo.ipynb`, Demo 1: angle-encoded state expansion.

Why angle encoding does not need vector normalization

In amplitude encoding, entries become amplitudes directly, so unit norm is a physical constraint.

In angle encoding, features are gate parameters:

$$x_i \mapsto R_\alpha(x_i).$$

Any real x_i defines a valid unitary, so no vector-level normalization is required for physical validity.

But scaling remains a modeling choice:

- periodicity can create collisions (aliasing)
- too large scales can make nearby points look almost random in angle space

What changes when we use phase rotations?

Compare

$$R_y(\theta)\,|0\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)|1\rangle$$

with

$$R_z(\theta)\,|0\rangle = e^{-i\theta/2}\,|0\rangle\,.$$

A pure R_z on $|0\rangle$ changes only a global phase, so by itself it does not change measurement probabilities.

After a Hadamard, phase is no longer global:

$$H\,|0\rangle \xrightarrow{R_z(\theta)} \frac{e^{-i\theta/2}\,|0\rangle + e^{i\theta/2}\,|1\rangle}{\sqrt{2}},$$

and relative phase now changes interference after later gates.

So phase encoding matters whenever the circuit creates superposition and interferes paths before measurement.

Pause B: solve Card Set B

Quantum feature maps

Feature map as a unitary

A quantum feature map is a data-dependent unitary:

$$|\phi(x)\rangle = U_\phi(x)\,|0\rangle^{\otimes n}.$$

Classical viewpoint:

$$x \mapsto \phi(x) \in \mathcal{F}.$$

Quantum viewpoint:

$$x \mapsto |\phi(x)\rangle \in \mathbb{C}^{2^n}.$$

The state itself is the lifted representation.

Terminology note: "feature map" means the map from raw inputs to a space where inner products become useful similarities. In QML this map is implicit in the state-preparation circuit.

Classical kernels versus quantum kernels

Classical kernel method: choose a functional family (RBF, polynomial, cosine, ...) and tune a few hyperparameters.

Quantum kernel method: choose a data-encoding unitary $U_\phi(x)$, then the kernel is induced:

$$K_Q(x, y) = \left|\langle 0|U_\phi(x)^\dagger U_\phi(y)|0\rangle|0\rangle|U_\phi(x)^\dagger U_\phi(y)|0\rangle\right|^2.$$

So we still "choose a kernel," but indirectly by choosing circuit structure (rotation family, entanglers, data re-uploading, depth, parameter scaling).

Quantum kernel and fidelity

For pure states $|\psi\rangle$ and $|\varphi\rangle$, the fidelity is

$$F(\psi, \varphi) = |\langle\psi|\varphi\rangle|^2 \in [0, 1].$$

Therefore a natural quantum kernel is

$$K_Q(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2.$$

Interpretation:

- $K_Q = 1$: encoded states are identical up to global phase
- $K_Q = 0$: encoded states are orthogonal
- intermediate values measure partial similarity

Fidelity here is exactly the squared overlap for pure states; this is the quantity estimated by adjoint trick and SWAP test later.

`lecture_05_demo.ipynb`, Demo 2: exact overlaps and fidelities.

One-qubit kernel: closed-form example

Choose

$$|\phi(x)\rangle = R_y(x)\,|0\rangle = \cos\left(\frac{x}{2}\right)|0\rangle + \sin\left(\frac{x}{2}\right)|1\rangle\,.$$

Then

$$\langle\phi(x)|\phi(y)\rangle = \cos\left(\frac{x-y}{2}\right),$$

and hence

$$K_Q(x,y) = \cos^2\left(\frac{x-y}{2}\right).$$

So the overlap is large for nearby angles and smaller for separated angles.

`lecture_05_demo.ipynb`, Demo 2: one-qubit closed-form kernel.

Euclidean, cosine, and overlap-based similarity

For vectors in $\mathbb{R}^d$:

$$d_E(x, y) = \|x - y\|, \qquad \cos(x, y) = \frac{x \cdot y}{\|x\|\|y\|}.$$

Typical use:

- Euclidean: magnitude-sensitive geometry
- cosine: direction-sensitive, scale-invariant geometry

Kernel trick needs a positive semidefinite kernel matrix. Both RBF from Euclidean distance and linear/cosine-style kernels can satisfy this with the right definition.

Quantum kernels are overlap kernels by construction, so they fit naturally into kernel-trick pipelines.

Classical distance, classical similarity, quantum overlap

For the same two inputs we can compare three notions:

$$\|x - y\|, \qquad K_{\mathrm{RBF}}(x, y) = e^{-\gamma\|x-y\|^2}, \qquad K_Q(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2.$$

They are not identical, but all three define geometry.

There is no universal ordering "distance $<$ classical kernel $<$ quantum kernel." Better means "better aligned with the task geometry," not "more powerful by definition."

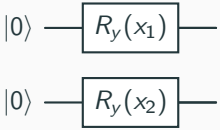
`lecture_05_demo.ipynb`, Demo 3: distance, classical similarity, and quantum fidelity.

Pause C: solve Card Set C

Entangling feature maps

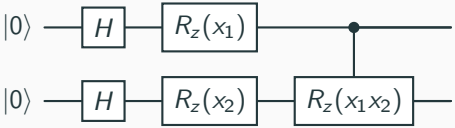
Product encoding versus entangling encoding

Product encoding:



Similarity factorizes strongly across qubits.

Entangling encoding:



Interaction terms enter the state geometry directly.

`lecture_05_demo.ipynb`, Demo 5: non-entangling versus entangling similarities.

ZZ-style interaction intuition

The Pauli matrix

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad Z \otimes Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

A ZZ-style phase block has the form

$$e^{-i\varphi(x_i, x_j)\, Z \otimes Z}.$$

This is a quantum analogue of introducing pairwise interaction terms, but it does so through phases and interference rather than by appending one extra classical coordinate.

In Qiskit this appears explicitly in `ZZFeatureMap`; the parameterized two-qubit phase angles control how strongly pairwise interactions influence the overlap geometry.

Is this analogous to XOR-type interactions?

Partial analogy: both add sensitivity to feature interactions.

Classical XOR/interaction lift:

$$(x_1, x_2) \mapsto (x_1, x_2, x_1 x_2).$$

ZZ feature map: interaction enters as phase through

$$e^{-i\varphi(x_1, x_2) Z \otimes Z}.$$

Similar goal, different mechanism: explicit extra coordinate versus phase-driven interference in Hilbert space.

Classical parallel

Classical lecture-4 story:

$$(x_1, x_2) \longmapsto (x_1, x_2, x_1 x_2).$$

Quantum lecture-5 story:

$$(x_1, x_2) \longmapsto U_\phi(x_1, x_2) \left|00\right\rangle ,$$

where an entangling block makes the overlap depend on joint feature effects.

In both cases we are enriching geometry without manually enumerating every useful nonlinear feature in the original input space.

Pause D: solve Card Set D

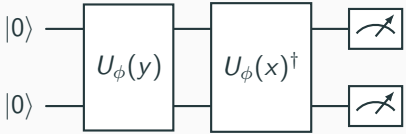
Estimating the kernel

Adjoint trick

For a feature map $|\phi(x)\rangle = U_\phi(x)\,|0\rangle^{\otimes n}$,

$$K_Q(x,y) = |\langle\phi(x)|\phi(y)\rangle|^2 = \left|\langle 0|\, U_\phi(x)^\dagger\, U_\phi(y)\,|0\rangle\right|^2.$$

Circuit picture:



The probability of measuring $0\ldots0$ is exactly the kernel value.

`lecture_05_demo.ipynb`, Demo 4: adjoint trick as a projection probability.

Why does $P(0\ldots0)$ equal the kernel?

Let

$$|\chi\rangle = U_\phi(x)^\dagger U_\phi(y)\,|0\rangle^{\otimes n}.$$

The Born rule gives

$$P(0\ldots0) = |\langle 0|\,\chi|^2 = \left|\langle 0|\,U_\phi(x)^\dagger U_\phi(y)\,|0\rangle\right|^2 = K_Q(x,y).$$

So the kernel is a projection probability onto the all-zero reference state.

Read this as:

1. prepare encoded $|\phi(y)\rangle$
2. "uncompute" with $U_\phi(x)^\dagger$
3. ask how much amplitude returned to $|0\ldots0\rangle$

The returned weight is exactly overlap squared.

SWAP test

Another route uses an ancilla qubit:

$$P(\mathrm{anc} = 0) = \frac{1+F}{2}, \qquad F = |\langle\phi(x)|\phi(y)\rangle|^2.$$

Operational picture:

1. prepare an ancilla in $|0\rangle$ and the two states $|\phi(x)\rangle$, $|\phi(y)\rangle$
2. apply Hadamard - controlled-SWAP - Hadamard on the ancilla
3. estimate $P(\mathrm{anc} = 0)$ and recover the fidelity

This is useful when we think operationally about fidelity estimation rather than directly composing $U_\phi(x)^\dagger U_\phi(y)$.

`lecture_05_demo.ipynb`, Demo 2 and Demo 4: fidelity and estimation routes.

Practical notes

Building a kernel matrix for N samples needs

$$\frac{N(N+1)}{2}$$

pairwise evaluations for a symmetric Gram matrix.

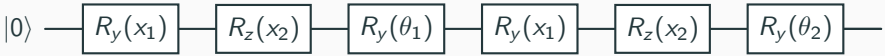
Practical caveats:

- shot noise perturbs the estimated similarities
- hardware noise changes overlaps and can wash out distinctions
- simulator studies are easier, but they do not remove the scaling cost

Pause E: solve Card Set E

Data re-uploading

Data re-uploading means encoding the same features more than once:



The idea is simple: data, trainable transformation, data again.

Why this can be richer:

- repeated non-commuting rotations generate more complex functions of the same input coordinates
- the induced kernel can include higher-order trigonometric interactions (Fourier-like harmonics), not just one basic frequency per feature

`lecture_05_demo.ipynb`, Demo 7: one-qubit re-uploading preview.

Re-uploading versus explicit classical feature engineering

Your intuition is correct: this is analogous in spirit to creating richer classical features (polynomials, interactions, transformed coordinates).

The difference is representational:

- classical: we explicitly construct new columns such as x_i^2, $x_i x_j$, $\log x_i$
- quantum re-uploading: we keep the same raw input coordinates, but repeated data-dependent unitaries implicitly generate a richer basis in the overlap kernel

So the model is not "only linear" in the raw input, even though each gate is a linear operator on the state vector.

Implementation workflow

Practical workflow:

1. choose an encoding circuit $U_\phi(x)$
2. compute pairwise similarities $K_{ij} = K_Q(x_i, x_j)$
3. compare against classical kernels under the same split
4. only then ask whether the quantum geometry helps

This lecture prepares the later quantum-kernel classifier material.

Transition to later lectures

Lecture 4 asked what a kernel is.

Lecture 5 asked how classical data becomes quantum states and how overlaps become kernels.

Later we will use these Gram matrices inside actual classifiers and compare them against strong classical baselines.